

Aditya Kulkarni

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SUMMARY

Full-stack engineer with production TypeScript/Node.js/React experience and a healthcare domain advantage. Shipped 10+ backend features across 7 deployments at Isomer AI (394 commits). Built ML-powered health tools at Mphasis (+25% claim accuracy), a compliance-ready RAG system for healthcare at Glance AI, and Sanjeevani (Top 1% of 1,100+, 87% ML accuracy). MS CS at NC State (May 2026). IEEE-published.

EXPERIENCE

Product Engineer Intern	Isomer AI	May 2025 - Present
- Developed 10+ backend features across 7 production deployments for an AI-powered claims automation platform; 394 commits in TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma, and Docker with real-time event pipelines (Pusher), CI/CD via GitHub Actions, and Sentry observability. - Built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 file storage), RBAC, and Auth0 authentication; implemented Temporal.io workflow orchestration and multi-model AI agent tooling (Claude, GPT-4o, Gemini) via MCP.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 - Aug 2024
Built a health claim processing automation tool using OCR, NLP, and ML classification models improving accuracy by 25%; automated CI/CD on AWS and GCP, integrated Grafana monitoring, and supported model validation and red-teaming.		
Graduate Research Assistant	NC State University	Apr 2025 - Present
Designed and deployed a cloud-native genomic data platform with JBrowse 2 on Google Cloud Run serving 500GB of datasets to 100+ researchers; improved ML pipeline accuracy by 12% via hyperparameter tuning and time-series analysis.		
AI/ML Intern	Capgemini	Jun 2023 - Aug 2023
Built distributed data pipelines using Hadoop and Apache Spark for batch processing and Apache Kafka streaming for real-time ingestion, reducing processing time by 15% and end-to-end latency by 20%.		

PROJECTS

Glance AI -- Privacy-First Healthcare RAG	Healthcare / AI
Compliance-ready RAG system for fintech and healthcare with per-tenant data isolation at the vector retrieval layer, preventing cross-tenant context leakage and LLM hallucinations across regulated environments; deterministic output architecture enforces factual correctness for healthcare workflows. Python, LLMs, Vector Databases, RAG, Privacy-First Architecture	
Sanjeevani -- Health App	Full Stack + ML
ML-powered health application achieving 87% accuracy using real-time sensor data and classification models; shortlisted Top 1% (12 of 1,100+) for innovation in UI and predictive health insights. Built full-stack with React integrating real-time data pipelines and health analytics dashboards. React, Python, ML Classification, Real-Time Sensor Data, Full-Stack	
PulseRank -- Real-Time Data Platform	Distributed Systems
CQRS event-driven platform: MySQL outbox + idempotency keys for exactly-once delivery, MongoDB TTL projections, Redis sorted sets for sub-ms reads; SSE live push; Testcontainers concurrency tests, k6 benchmarks (10K users), OpenTelemetry tracing, Prometheus + Grafana observability. TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry	
VectorLift -- Semantic Search Engine	Search / ML
Neural IR pipeline on MS MARCO: BM25 sparse baseline, dense bi-encoder retrieval (sentence-transformers, in-batch negatives), cross-encoder re-ranker; Kafka-driven indexing into Elasticsearch + Qdrant; FastAPI serving with per-stage latency tracing; NDCG@10/MRR@10 evaluation. Python, PyTorch, Elasticsearch, Qdrant, Kafka, FastAPI, Prometheus, Grafana, Docker	

EDUCATION

North Carolina State University -- MS Computer Science	Aug 2024 - May 2026
Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University -- BTech Computer Science CGPA: 9.32/10	Aug 2020 - May 2024
Machine Learning, Big Data Infrastructure, DS & Algorithms, Blockchain	

SKILLS & AWARDS

Languages: TypeScript, JavaScript, Python, Java, SQL, C | **Frontend:** React, Next.js, Tailwind CSS | **Backend:** Node.js, Express, NestJS, FastAPI | **Databases:** MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch | **Cloud / DevOps:** AWS, GCP, Docker, Kubernetes, CI/CD, Terraform | **AI/ML:** LangChain, PyTorch, Hugging Face, RAG, OpenAI/Anthropic/Google SDKs

HackNC24 Top 10 (100+ teams, 48h) | **Sanjeevani Top 1%** (12 of 1,100+) | **IEEE Published** -- Mapify (500+ attendees) | **Google DSC Lead** (500+ students)